# Agent-based Diamond–Dybvig with cultural heterogeneity

Walter Stover · George Mason University · 26 October 2026

# The puzzle: who runs?




*CryptoSlate, 12 October 2026, on the Financial Stability Board's note to the G20. Screenshot, cryptoslate.com.*

# Who runs? Luck of the draw, and location

## Variation in the signal

- Goldstein–Pauzner: every depositor gets a noisy draw of the same signal
- The draw breaks common knowledge and selects the equilibrium
- Who runs: whoever drew low
- Everyone decides at once, so there is no first

## Variation in location

- Networks and media decide who hears what, from whom, when
- Kelly–Ó Gráda, Iyer–Puri, Cookson et al.
- Who runs first: whoever sits nearest the source

# Variation in interpretation

## When there are two signals, which signal does a depositor act on, and in what proportion?

**Latent in the canonical models.** Diamond–Dybvig has no signal; Morris–Shin and Goldstein–Pauzner have one. The question cannot arise.

**Askable once a public signal is added.** Morris–Shin (2002), Hellwig (2002), Angeletos–Werning (2006) put a public signal next to the private one.

**The question a panic poses.** A depositor in a run holds two things that disagree: what they see and what they are told.

# What panics look like

**Panics are open disagreements between the public line and private evidence, and depositors split on which to trust.**

**Celsius, 2022.** Official posts insisting the platform was sound, while CEL fell and withdrawals accelerated.

**Silicon Valley Bank, 2023.** A call asking clients to stay calm, against a Twitter feed saying the opposite.

**Northern Rock, 2007.** A Bank of England support announcement, against queues outside the branches.

*In each case the interesting fact is who moved on which signal.*

# Cultural economics: a source of stable weights

**Culture supplies a weight that differs across depositors, is set before the run, and does not move with the information environment.**

**Transmitted, not chosen.** Cultural values pass largely unchanged from parents to children (Guiso, Sapienza, Zingales; Bazzi et al.). The weight is a disposition, not a calculation.

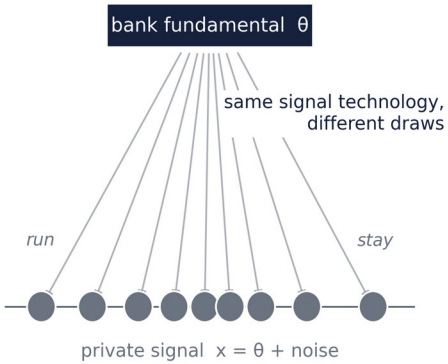
**Individualism is the right dimension.** Self-reliance against deference to the group (Hofstede) is the private-against-social weighting the model already contains (Gorodnichenko–Roland).

**Measurable per depositor.** Given names carry it (Twenge et al.), so the weight can be observed, not assumed.

# The ladder: signal, location, interpretation

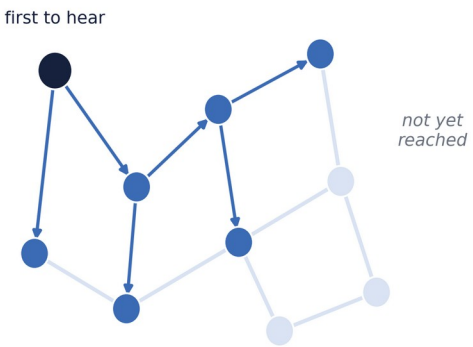
**1 Signal**

one fundamental, one noisy draw each




**2 Location**

who hears it, from whom, when

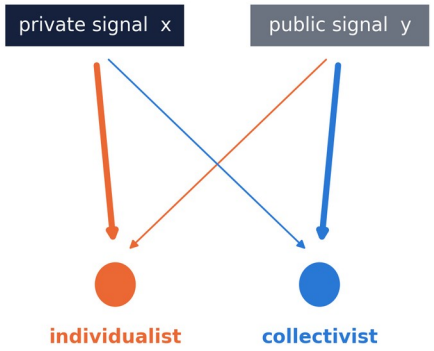



networks and media: Kelly–Ó Gráda, Iyer–Puri, Cookson et al.

**3 Interpretation**

the same two signals, different weights

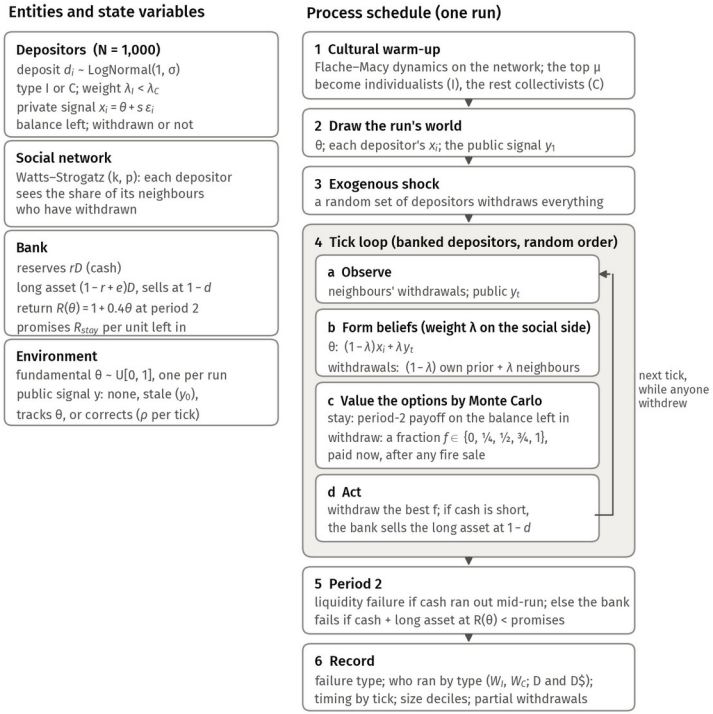
*culture sets the weight*




leans on what it sees  leans on the public line

Rungs 1 and 2 are the literature. **Rung 3 is this paper:** the only one that orders depositors by a trait, and reverses when the public line is too optimistic or too pessimistic.

# The model in one picture




**1,000**
depositors on a small-world network, one bank

**1%**
cap on any one deposit; sizes log-normal (σ 2.5) below it

**CARA 0.1**
risk aversion; partial withdrawals allowed

**250**
runs per cell: 25 initialisations × 10 draws

- One bank that sells its long asset at a discount when cash runs short (a fire-sale bank)
- Each tick a depositor sees neighbours' withdrawals and the public news, blends them with a private signal, and withdraws if that pays
- A public signal can be absent, accurate, or outdated
- Outcomes: D = individualists' minus collectivists' withdrawal share (those not hit by the shock); P(fail) = share of runs that fail

# Individualists and collectivists: who listens to whom

## Individualists

- Trust their own information about the bank
- Put little weight on what neighbours do or on public news
- Weight on the social side: $\lambda = 0.1$ to $0.3$
- Act on what they see themselves

## Collectivists

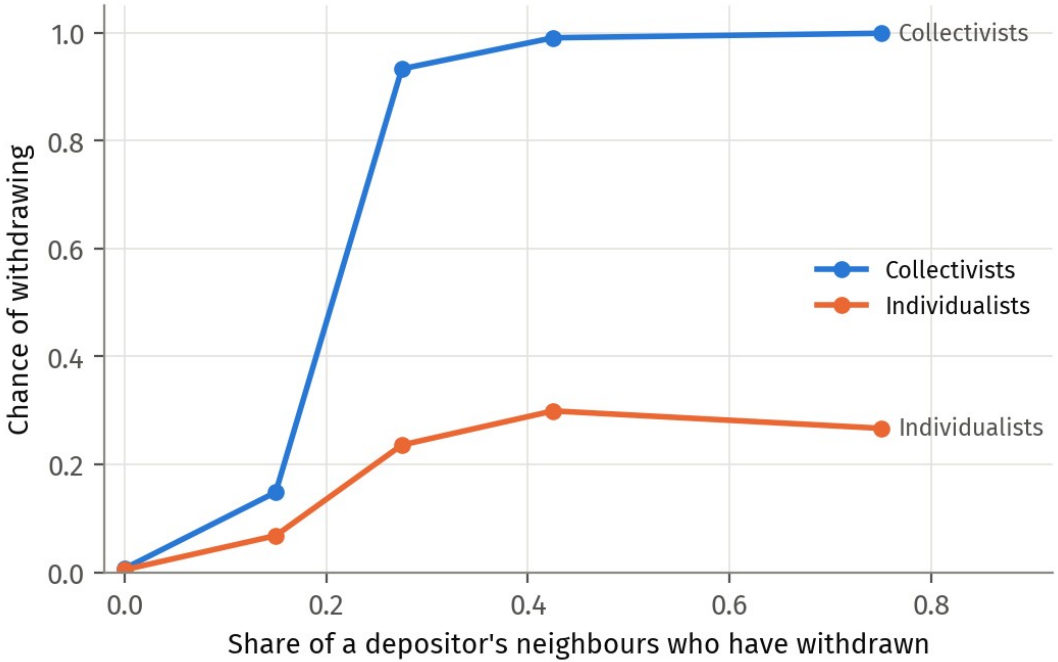
- Watch what neighbours do and what the public news says
- Put little weight on their own information
- Weight on the social side: $\lambda = 0.5$ to $0.9$
- Move when others move

**Three pairs ($\lambda$ individualist, $\lambda$ collectivist) in the figures**

**(0.1, 0.9)** wide gap **(0.2, 0.8)** medium **(0.3, 0.5)** narrow

The share of individualists $\mu$ is set by a cultural warm-up on the network. **Orange = individualists**, **blue = collectivists** throughout.

# Collectivists herd




*Chance of withdrawing, given the share of a depositor's neighbours who have already withdrawn. λ pairs pooled; reference bank.*

**0.15 to 0.93**

collectivists, once 20–35% of their neighbours are out

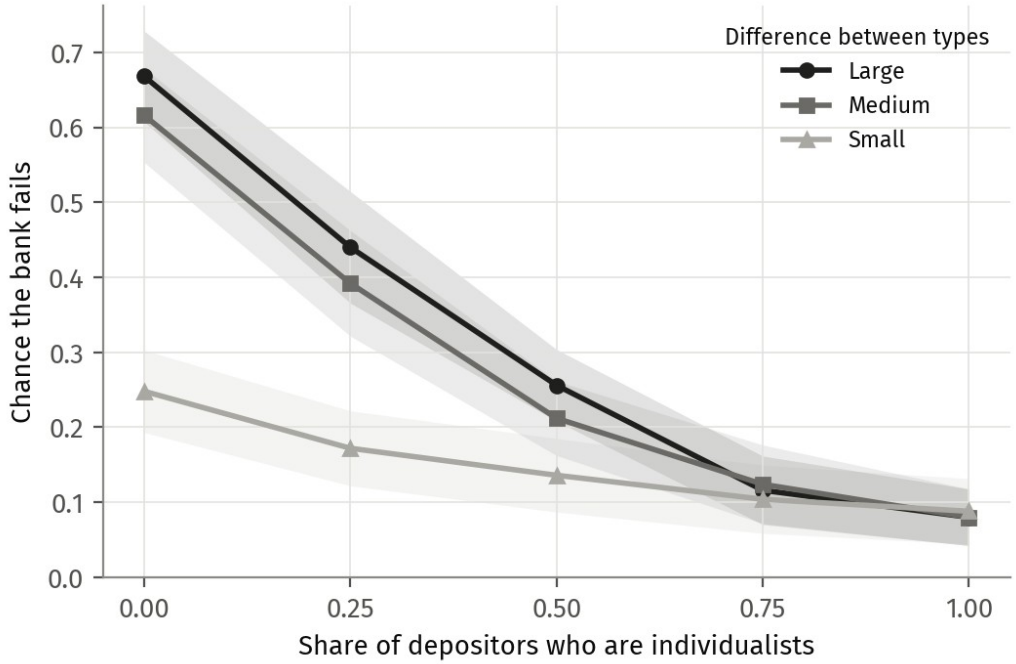
**0.25 to 0.30**

where individualists level off, whatever their neighbours do

**+0.58**

difference in slopes (collectivists 0.80, individualists 0.22), t = 16.7

# More individualists, fewer failures




*Chance the bank fails, by share of individualists μ, one line per λ pair. Includes the pure populations (μ 0 and 1).*

**0.67 to 0.08**

P(fail) from no individualists to all, wide λ gap

**0.25 to 0.09**

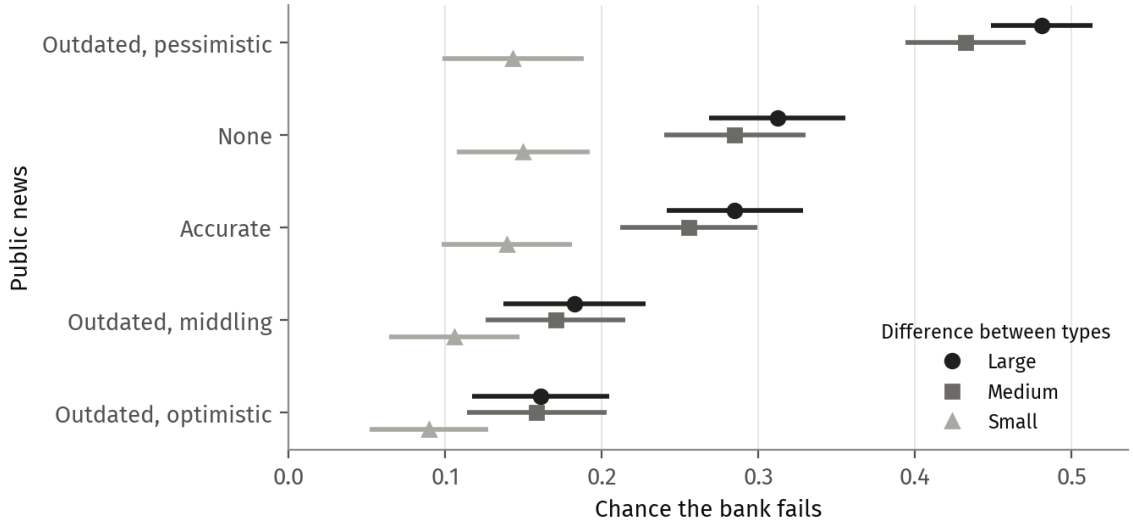
the same, narrow λ gap: it matters most when collectivists dominate

**Monotone**

No interior peak at any mix. The lines converge once individualists are 75% or more.

Agent-based Diamond–Dybvig with cultural heterogeneity

# Outdated news cuts both ways




*Chance the bank fails, by the kind of public news, dot and CI per λ pair. All mixes μ. Numbers at right: wide λ gap.*

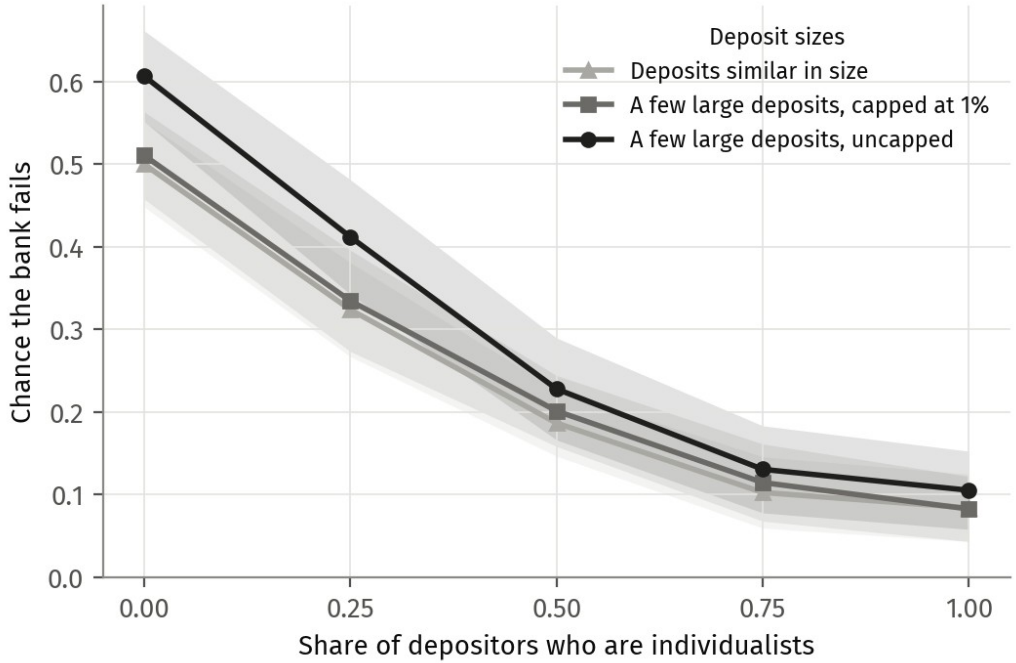
**0.48**
outdated, pessimistic news

**0.31**
no public news

**0.29**
accurate news

**0.16**
outdated, optimistic news

# Capping the largest deposits




*Chance the bank fails by share of individualists, for three deposit-size distributions. λ pairs pooled.*

**0.61 to 0.11**

uncapped: a few very large deposits (σ 2.5)

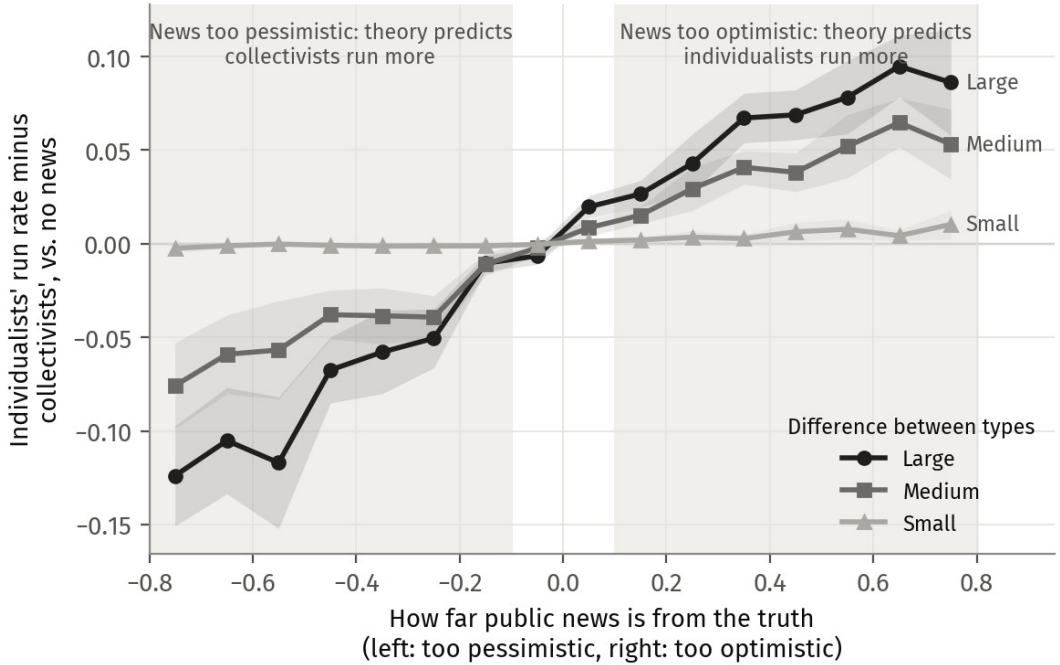
**0.51 to 0.08**

capped: no deposit above 1% of the total (the reference bank)

**0.50 to 0.08**

similar sizes (σ 1). The μ gradient is the same in all three.

# Does the model hold up?




*D relative to no news, against how far the public news sits from the truth. Shaded: where the theory predicts the sign.*

### Pre-registered tests

G1′ optimistic news: individualists run more +0.033

G4′ pessimistic news: collectivists run more −0.031

G2a′ the effect grows with the λ gap +0.093

G3 sorting with no news: counts against (one pair wrong-signed)

**3 of 4**

confirmatory tests support the theory, on all three versions of the bank

# Takeaways and next steps

| | | |
|---|---|---|
| 1 | **Sorting follows the news** | Optimistic public news makes individualists run relatively more; pessimistic news, collectivists. 3 of 4 pre-registered tests pass, in all three bank versions. |
| 2 | **Collectivists herd** | Their chance of withdrawing jumps from 0.15 to 0.93 once 20–35% of neighbours are out; individualists level off near 0.3. |
| 3 | **More individualists, fewer failures** | P(fail) falls from 0.67 to 0.08 as the individualist share goes from 0 to 1. Monotone; no interior peak. |
| 4 | **Stale news cuts both ways** | A stale optimistic signal halves failure, a stale pessimistic one raises it by half. Accurate news barely moves it. |
| 5 | **Capping the largest deposits** | A 1% cap brings failure down to the no-concentration bank's level; the culture gradient is unchanged. |
| 6 | **Not yet matched** | How Celsius's largest depositors exit: one-shot share 0.35 there, 0.93 or more in the model. |

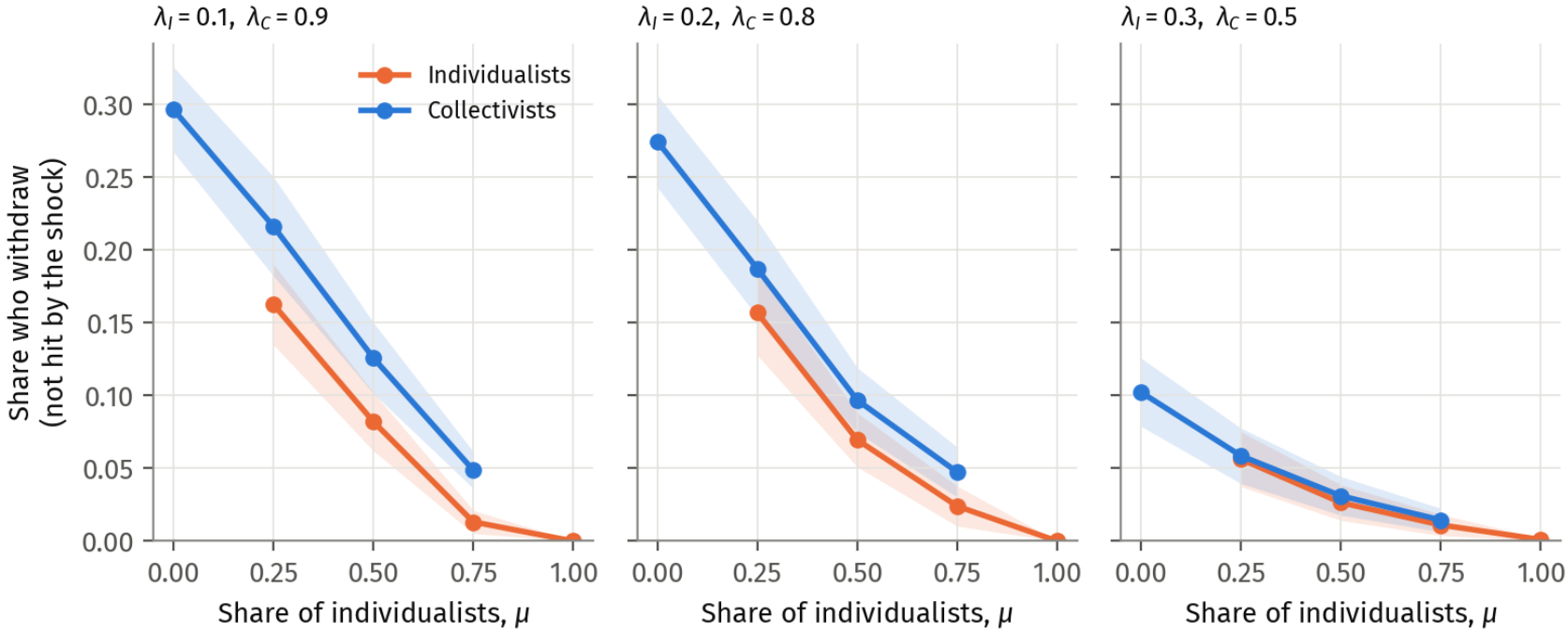
## Next steps

- Super-individuals, to match how the largest depositors exited Celsius
- Deposit insurance and the reserve ratio on the fire-sale bank
- Network topology and learning
- Full sweep: stale public signal plus no-signal baseline, CARA 0.1, σ reported

# Backup

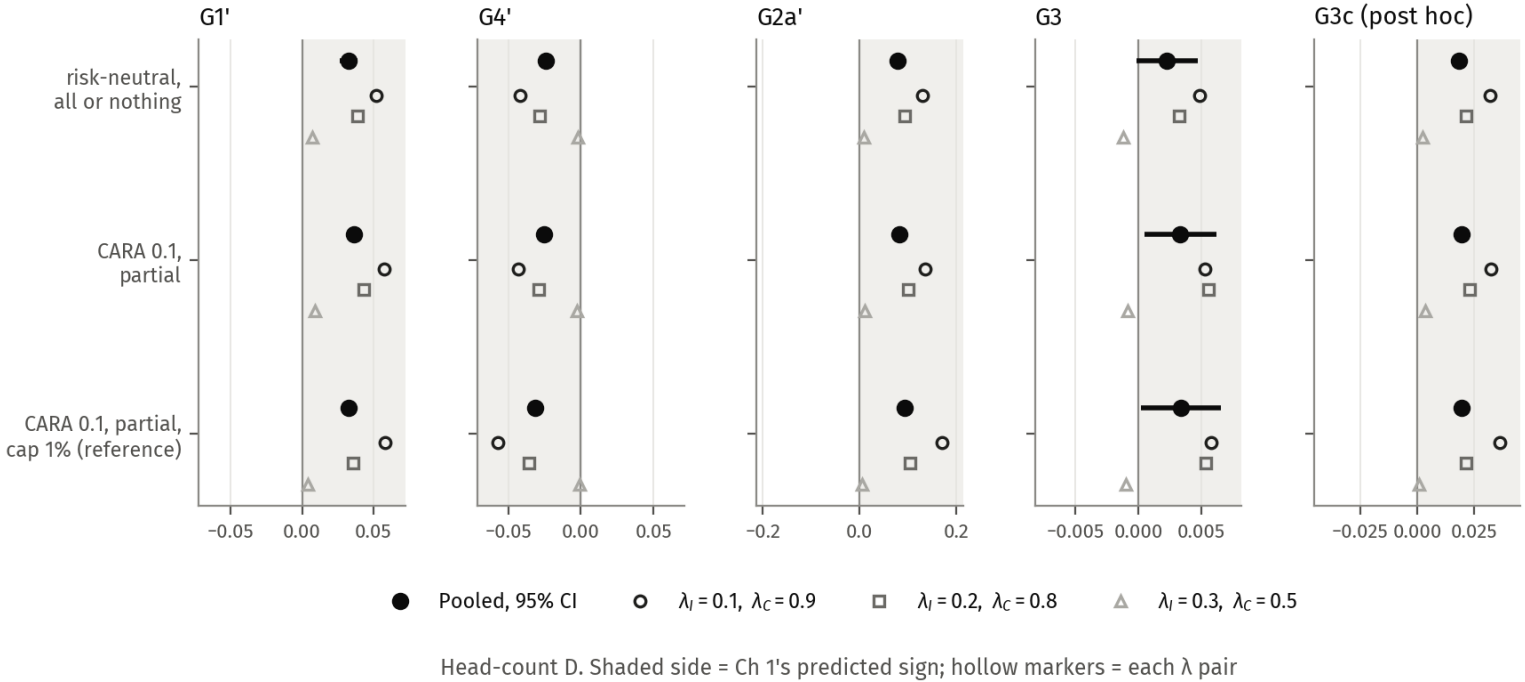
Technical versions of the exhibits, the pre-registered tests, channels, and the Celsius docking gap.

# Who withdraws, by type and mix



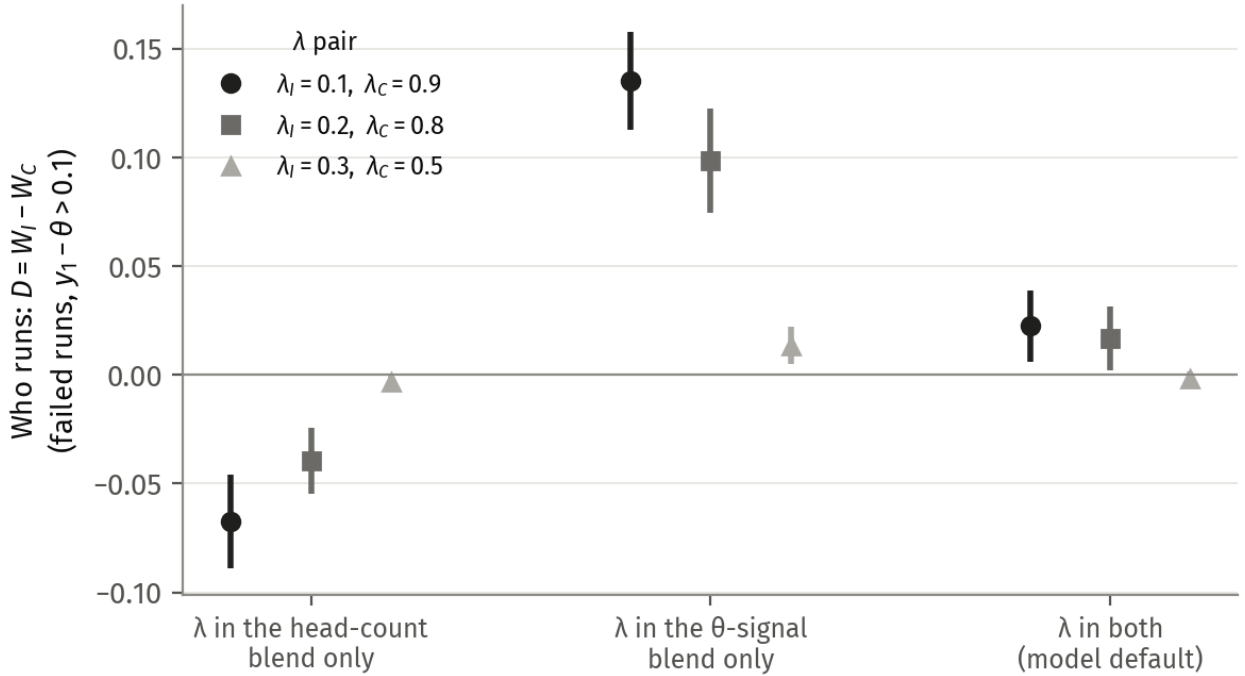

*Share who withdraw by individualist share μ, one panel per λ pair; orange individualists, blue collectivists, with CIs.*

# The pre-registered tests, three versions of the bank



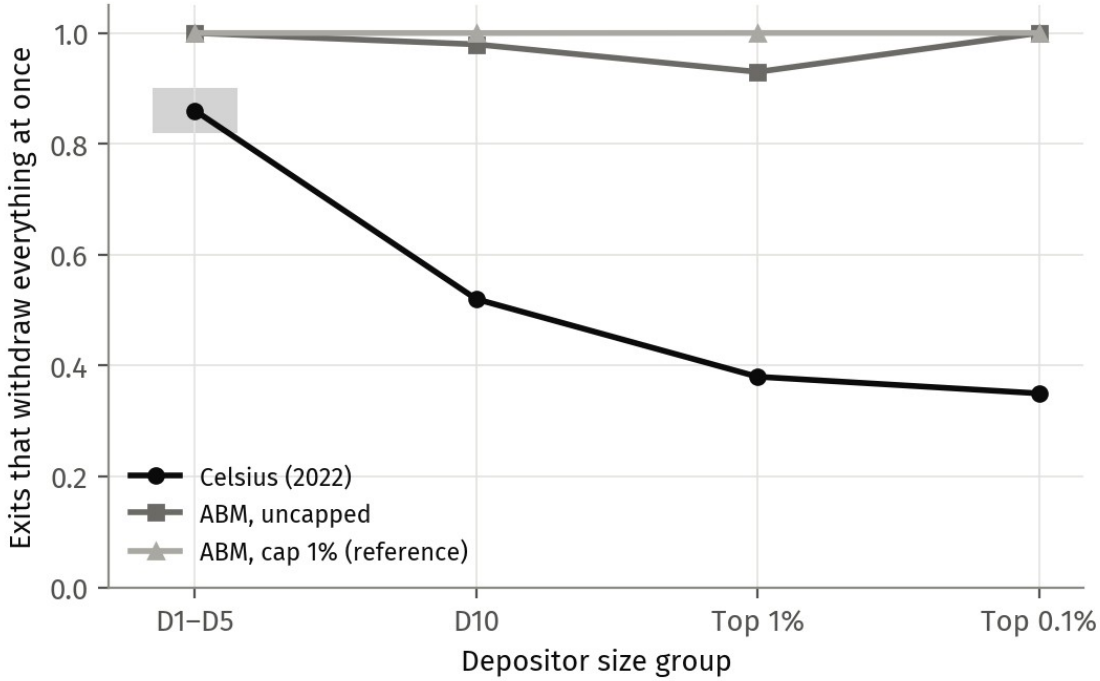

*G1′, G4′, G2a′, G3 and G3c (post hoc). Rows: risk-neutral all-or-nothing, CARA partial, capped reference bank. Filled dot: pooled; hollow: each λ pair. Shaded: the theory's predicted sign.*

# Which channel carries culture



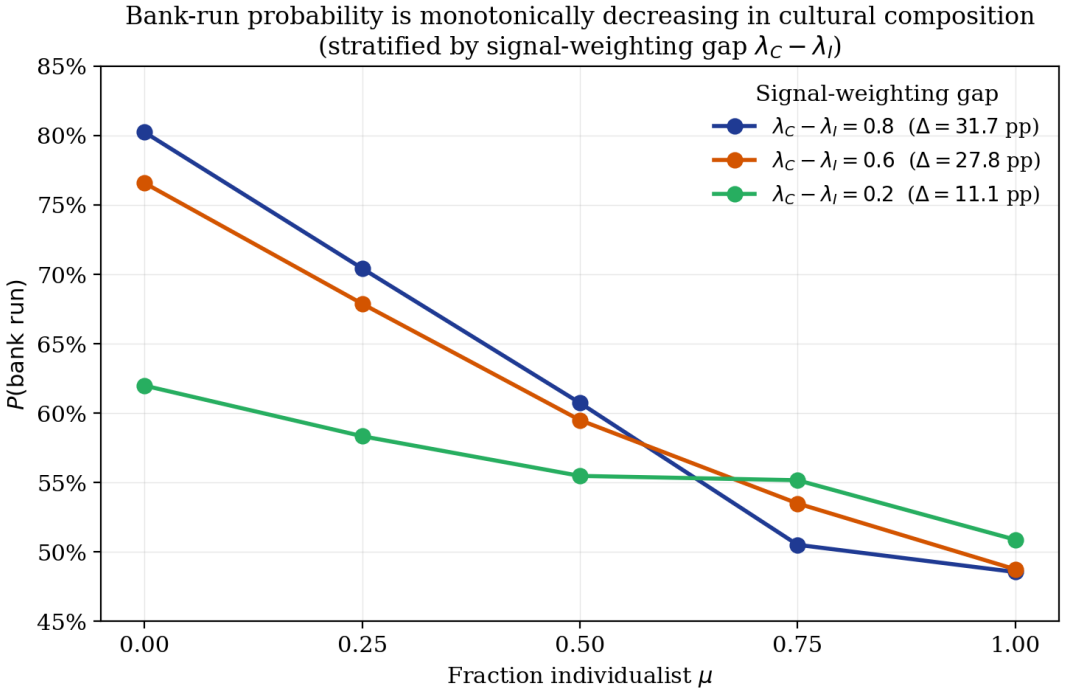

*D in failed runs with optimistic news, when λ weights only the neighbour blend, only the news blend, or both (the model). θ-blend only +0.14; head-count only −0.07; both +0.02.*

# Matching Celsius: how large depositors exit




*Share of exits that withdraw everything at once, by depositor size. Black: Celsius (band for D1–D5); greys: the model, uncapped and capped.*

# Earlier specification (vault bank): failure by mix




*Vault bank, earlier specification (chapter Figure 1): probability of a run by share of individualists, three λ gaps. Also monotone.*